

Gourilakshmi Neerajkumar

neerajk2@uwm.edu | www.linkedin.com/in/gourilakshmineerajkumar | <https://s4ig0n.github.io/Portfolio/>

Education

University of Wisconsin- Milwaukee

May 2027

B.S Computer Science (with Honors), Minor in Information Sciences and Technology, Microcredential in AI

Dean's list student | 3.7 GPA

Skills

Programming: C, C++, Python, Java, MATLAB, Django, SQL/MySQL, HTML/CSS, Assembly, Bootstrap

Tools/Tech: Arduino, Raspberry Pi, MPLAB X, Git, Linux, Docker, QEMU, MQTT, Jenkins, Bitbucket, SonarQube, Jira, TestRail, LaTeX

Experience

Quality Assurance Intern

May 2026 - Aug 2026

Mercury Security-HID Global

- Developed a cross-architecture MQTT file transfer pipeline between an x64 Windows host and an ARM-based MP series Linux controller, using Docker and QEMU for deployment, JSON messages for file metadata, enabling the controller to bundle requested files into a single tar archive that downloads automatically to the PC.
- Validated 2 full release targets and 2 hotfix targets for access control controllers by executing TestRail test runs on Jenkins builds and writing targeted test cases to verify Jira ticket fixes
- Resolved Jira bug tickets across controller firmware and its web-based configuration interface, merging fixes to master in Bitbucket after peer review and validating builds through Jenkins and SonarQube
- Configured wired and wireless readers and locks and ran compatibility testing across the company's full lineup of SIOs, readers, and locks, documenting results in an extensive test matrix

Web Developer – Mothers Against Gun Violence

Jan 2026 – May 2026

Nonprofit client project, Milwaukee (mentored by Direct Supply)

- Redesigned a Milwaukee nonprofit's website with a four-person team, applying a new brand identity and color palette
- Streamlined online giving by integrating a donation plugin, simplifying the donation process for supporters
- Expanded site content with new pages covering the founder's story and a map of murals across Milwaukee honoring gun violence victims, while improving site accessibility

Exhibit Maintenance- Discovery World

July 2024 – Sep 2024

Discovery World, Milwaukee

- Redesigned and implemented a 3D circuit for the Wimshurst machine using an Arduino Uno and multiple relays programmed in C++
- Rebuilt the Elements Display to increase visitor interactions by using Raspberry Pi (Python) and touch pads.

Research & Projects

Undergraduate Researcher – Graph Theory Project

May 2025 – Present

Department of Mathematical Sciences, University of Wisconsin–Milwaukee, MAA, NSF

- Conducted research on Truncated Square Graphs focusing on *Total Restricted Broadcast Domination (TR Broadcast Domination)* aiming to answer questions proposed by J. Cervantes and P. Harris.

Scheduling Management Web App (Team Project) | Full-Stack Developer

Feb 2025 – May 2025

- Developed a role-based scheduling system with dashboards for students, TAs, instructors, and admins
- Built backend functionality in Django and designed responsive UI components using Bootstrap.

Support for Undergraduate Research Fellows (SURF), Two-Time Recipient

Sep 2023 – Dec 2024

College of Engineering & Applied Sciences, University of Wisconsin–Milwaukee

- Advanced a solar sculpture project that harnesses sunlight by day to power visual performances at night, using a PIC24 microcontroller and circuit debugger in MPLAB X on Linux
- Tested and debugged an LC-3 visualization tool and wrote its project documentation and user manual

Leadership & Involvement

Teaching Assistant – Girls Who Code, Milwaukee

Sep 2026 – Present

- Facilitate hands-on Python instruction for elementary and middle school girls in the Milwaukee area

President – Student Association of India, UW–Milwaukee

Aug 2024- Present

- Coordinated over 20 cultural and social events with over 250 attendees celebrating Indian traditions at UWM

Math Tutor – Student Success Center, UW–Milwaukee

Aug 2024 – Present

- Helped over 300 students work through challenging math concepts by breaking them down step by step